THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 5: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Parallel chords of lengths 14 cm and 48 cm lie on opposite sides of the centre of a circle of radius 25 cm. The distance between them (in cm) is: 1

(a) 33 **(b)** 31 **(c)** 30 **(d)** 32

2. Parallel chords of lengths 14 cm and 48 cm lie on the same side of the centre of a circle of radius 25 cm. The distance between them (in cm) is: 1

(a) 16 **(b)** 18 **(c)** 19 **(d)** 17

3. Two circles of radii 13 cm and 15 cm intersect in a common chord of length 24 cm and their centres lie on opposite sides of it. The distance between the centres (in cm) is: 1

(a) 13 **(b)** 14 **(c)** 15 **(d)** 16

4. $ABCD$ is cyclic, $\angle ABD = 35^\circ$ and $\angle DBC = 25^\circ$. Then $\angle ABC =$ 1

(a) 70° **(b)** 50° **(c)** 60° **(d)** 80°

5. The exterior angle at D of cyclic quadrilateral $ABCD$ (side CD produced) equals the interior angle at B, which is 72°. The interior angle $\angle ADC$ is: 1

(a) 118° **(b)** 98° **(c)** 128° **(d)** 108°

6. AB is a diameter, C and D are on the circle (same side of AB) and $ABCD$ is cyclic with $\angle DAB = 48^\circ$. Then $\angle BCD =$ 1

(a) 132° **(b)** 142° **(c)** 152° **(d)** 122°

7. A chord AB of length 14 cm lies 24 cm from the centre O. The area of triangle OAB (cm^2) is: 1

(a) 170 **(b)** 169 **(c)** 167 **(d)** 168

8. If two chords of a circle bisect each other, they are: 1

(a) both diameters **(b)** unequal

(c) perpendicular to each other **(d)** parallel

9. **Assertion (A):** In a circle of radius 5 cm, a chord at a distance of 3 cm from the centre has length 8 cm. 1
Reason (R): The perpendicular from the centre of a circle to a chord bisects the chord.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** Equal chords of a circle are equidistant from the centre. 1
Reason (R): The perpendicular from the centre of a circle to a chord bisects the chord.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Two parallel chords of lengths 14 cm and 48 cm lie on the opposite sides of the centre of a circle of radius 25 cm. Find the distance between them. **2**

12. Two circles of radii 10 cm and 17 cm intersect, and the common chord is 16 cm. Find the distance between the centres if they lie on opposite sides of the chord. **2**

13. $ABCD$ is a cyclic quadrilateral and $\angle ABD = 32^\circ$, $\angle DBC = 41^\circ$. Find $\angle ABC$ and $\angle ADC$. **2**

14. Two parallel chords of lengths 30 cm and 40 cm lie on the opposite sides of the centre of a circle of radius 25 cm. Find the distance between them. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Two circles of radii 13 cm and 15 cm intersect in a common chord of length 24 cm. Find the distance between their centres if they are on **3**
(i) opposite sides,
(ii) the same side of the common chord.

16. Parallel chords of lengths 10 cm and 24 cm in a circle of radius 13 cm. Find the distance between them in both possible positions. **3**

17. AB is a diameter of a circle; C, D lie on the circle on the same side of AB so that $ABCD$ is a cyclic quadrilateral with $\angle DAB = 34^\circ$. Find $\angle BCD$, $\angle ADB$ and $\angle ABD$. **3**

OR

Two circles of radii 13 cm and 20 cm intersect in a common chord of length 24 cm. Find the distance between their centres if they are on **3**
(i) opposite sides,
(ii) the same side of the common chord.

18. Parallel chords of lengths 16 cm and 30 cm in a circle of radius 17 cm. Find the distance between them in both possible positions. **3**

OR

AB is a diameter of a circle; C, D lie on the circle on the same side of AB so that $ABCD$ is a cyclic quadrilateral with $\angle DAB = 41^\circ$. Find $\angle BCD$, $\angle ADB$ and $\angle ABD$. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A circular garden in a Chandigarh park has radius 13 m. A straight path, 24 m long, joins two points on its boundary (a chord of the circle). **5**
(i) Find the half-length of the chord.
(ii) Find the perpendicular distance of the path from the centre.
(iii) Find the length of another chord which is at a distance of 12 m from the centre.
(iv) If the two chords lie on opposite sides of the centre and are parallel, find the distance between them.

20. In a cyclic quadrilateral $ABCD$, $\angle A = (3x + 5)^\circ$, $\angle B = (3y + 6)^\circ$, $\angle C = (2x + 15)^\circ$ and $\angle D = (3y + 12)^\circ$. Find x, y and all four angles. **5**

OR

A circle of radius 15 cm has a chord AB of length 18 cm. Find the distance of AB from the centre O, the area of $\triangle OAB$ and the perimeter of $\triangle OAB$. **5**

END OF QUESTION PAPER — Total: 40 Marks